

Consider the system of linear equations

$$x + y - z = 2$$

$$3x + 2y - z = 3$$

$$-x - y + 2z = -1$$

Write down the system in matrix form,

$$AX = b \text{ where } X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Write down the matrices A and b as well as their respective sizes:

- ☐ $A = \begin{bmatrix} 1 & 1 & -1 \\ 3 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$, A is a 3 by 3 matrix and b is a 3 by 1 matrix
- ☒ $A = \begin{bmatrix} 1 & 1 & -1 \\ 3 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$, A is a 3 by 3 matrix and b is a 3 by 1 matrix
- ☐ $A = \begin{bmatrix} 1 & 1 & -1 \\ 3 & 2 & -1 \\ -1 & -1 & -2 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$, A is a 3 by 3 matrix and b is a 3 by 1 matrix
- ☐ None of the given options are correct.
- ☐ $A = \begin{bmatrix} 1 & 1 & -1 \\ 3 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$ and $b = \begin{bmatrix} -2 \\ 3 \\ -1 \end{bmatrix}$, A is a 3 by 3 matrix and b is a 3 by 1 matrix

Clear my choice